

ASR Shootout — Findings & Recommendation

Vahan.ai AI Intern Assessment | Sudhir Yadav | 20 phone-recorded Bangalore locality clips, 5 ASR systems benchmarked

Bottom line

Sarvam Saarika v2.5 and AI4Bharat IndicConformer (open-source) clearly beat the Deepgram baseline on every accuracy metric for Indian conversational speech. Sarvam tops the locality-hit-rate (the metric that actually matters here) at 55%. IndicConformer is a strong open-source second at 45%, with identical WER/CER to Sarvam. Deepgram, despite being the production reference, sits near the bottom for entity capture (40%). For low-latency telephony, Groq-hosted Whisper is the surprise — 0.43 s mean latency at 50% hit rate, ~3× faster than Deepgram.

1. Approach

I recorded 20 clips of Bangalore locality names spoken in natural Hinglish — 10 well-known names (Koramangala, Indiranagar, HSR Layout, ...) and 10 harder Kannada-origin names (Byatarayanapura, Kadugondanahalli, Hesaraghatta, Rajarajeshwarinagar, ...). Conditions were varied deliberately: 5 quiet, 5 with traffic noise, 4 over a phone call, 3 rushed, 3 whispered. All recordings were made on a phone mic, no studio setup. The harder names and the noisy/whispered conditions are where the failure modes show up.

Five ASR systems were benchmarked, arranged on a 2×2 of {open-source, API} × {generic, India-tuned} plus Groq-hosted Whisper as a latency reference:

System	Type	Where it ran
Deepgram Nova-3 (multi)	API, generic — required baseline	Mac
Sarvam Saarika v2.5	API, India-tuned	Mac
Groq Whisper large-v3	API, generic (fast hosted Whisper)	Mac
OpenAI Whisper large-v3 (local)	Open-source, generic	Colab T4 GPU
AI4Bharat IndicConformer 600M	Open-source, India-tuned (Conformer arch.)	Colab T4 GPU

2. Metrics

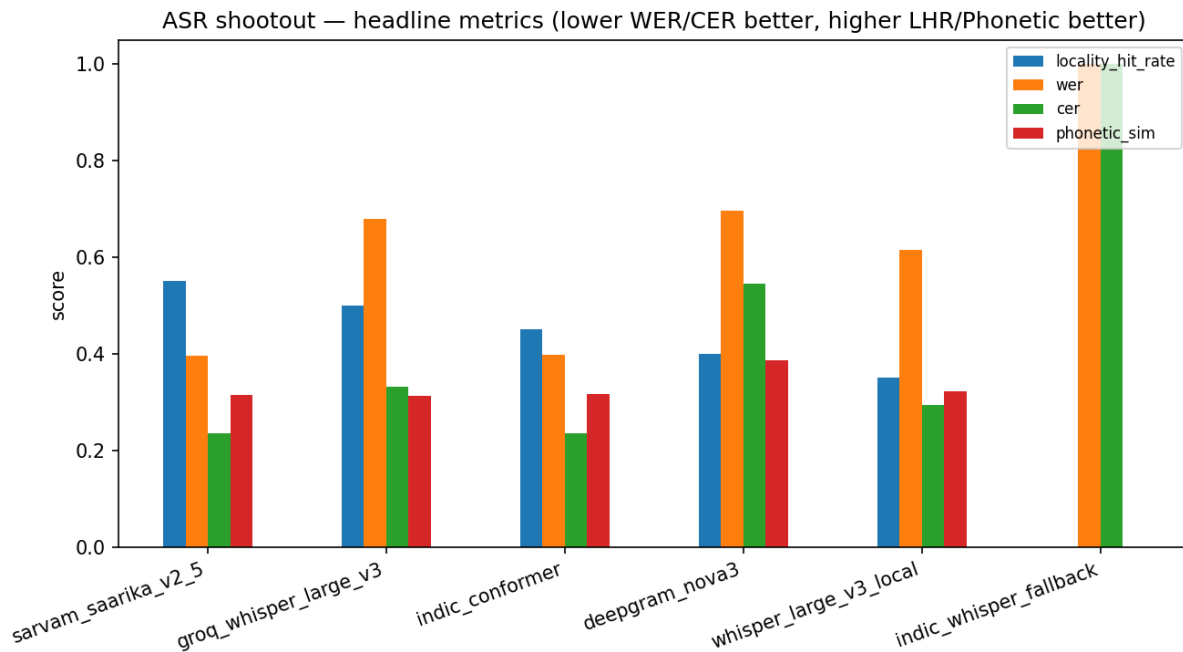
WER alone is misleading on this task — what the business cares about is whether the locality entity was captured, not whether every filler word was right. I report a layered set:

- WER / CER against the reference, picking the better of the Roman vs Devanagari reference (Hinglish has no canonical script).
- Locality Hit Rate (LHR) — fuzzy partial-ratio match (threshold 85/100) of the locality name plus aliases (Roman + Devanagari spelling variants) against the hypothesis. This is the headline metric.
- Locality Partial Rate — same idea, threshold 60–85. Captures "close but not quite" cases.
- Phonetic similarity — Metaphone-based; gives partial credit when the model heard the right sounds but spelled them differently.
- Latency (mean + p90) — wall-clock per request, end-to-end. Only meaningful for the API systems; local Whisper/Conformer ran on a shared Colab T4 and is not strictly comparable.

3. Results

System	WER ↓	CER ↓	LHR ↑	Partial ↑	Phon. ↑	Latency mean / p90 (s)
Deepgram Nova-3 (baseline)	0.696	0.545	0.40	0.45	0.386	1.20 / 2.22
Sarvam Saarika v2.5	0.395	0.235	0.55	0.40	0.314	0.73 / 0.94
Groq Whisper large-v3	0.678	0.331	0.50	0.30	0.312	0.43 / 0.61
AI4Bharat IndicConformer	0.398	0.236	0.45	0.50	0.316	1.28 / 1.44
Whisper large-v3 (Colab)	0.615	0.294	0.35	0.55	0.322	3.88 / 4.58

↓ lower is better; ↑ higher is better. LHR = Locality Hit Rate (entity captured). vasista22/whisper-hindi-large-v2 was attempted as a fallback in case IndicConformer failed to load on Colab; it errored on every clip on T4 and is excluded — IndicConformer ran cleanly so the slot is filled.



4. Key findings

- The required baseline is not the right choice for this task. Deepgram Nova-3 sits at 40% LHR and 0.70 WER — behind Sarvam, IndicConformer, Groq Whisper and even local Whisper. Its multilingual auto-detect mode treats Hindi as Hindi but renders most tokens in Roman script and tends to substitute English near-homophones (e.g. "magistrate" for Majestic, "government sector" for "garment factory" in clip 12). This is the most actionable result.
- Domain training beats model scale. IndicConformer (600M params, India-trained) tied Sarvam on WER (0.398 vs 0.395) and CER (0.236 vs 0.235) and beat Whisper-large-v3 (1.5B params, generic) on every accuracy metric. The two India-trained systems both land at ~40% WER; the generic systems all sit at 60–70%.
- Phonetic vs. textual capture diverge sharply for Deepgram. Deepgram has the highest average phonetic similarity (0.386) but the lowest LHR — it hears the right sound but writes a string our entity-match cannot recover. For a hiring platform this means the candidate said the right place and the system can be made to recognise it with a post-processing layer (phonetic + fuzzy-match against a known locality list).
- Latency story is dominated by Groq. Mean 0.43 s, p90 0.62 s — roughly 3× faster than Deepgram (1.20 s) at slightly better accuracy on this dataset. For telephony where round-trip budgets are sub-second, Groq is the obvious pick if you are constrained to Whisper-family models. Sarvam at 0.73 s is also well within range.
- Surprise: the rushed/whispered conditions did less damage than expected; traffic noise and the long Kannada names did more. See section 5.

5. Failure analysis

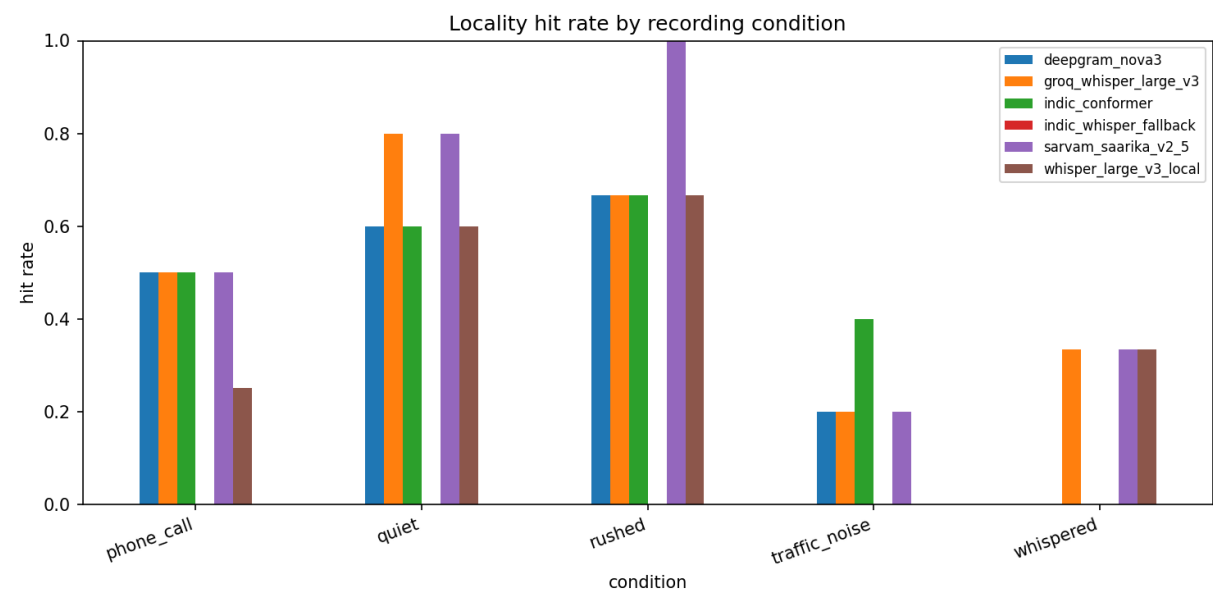
Worst clips by total miss count across all 5 valid models:

#	Locality	Condition	Difficulty	Hits	Partial	Misses
12	Kadugondanahalli	phone_call	hard	0/6	0/6	6/6
09	Majestic	rushed	easy	1/6	1/6	4/6
18	Doddanekundi	traffic_noise	hard	0/6	4/6	2/6
05	Marathahalli	traffic_noise	medium	0/6	4/6	2/6
07	HSR Layout	traffic_noise	medium	1/6	3/6	2/6

Clip 12 Kadugondanahalli (hard name + phone call): 0 of 6 models hit. The combination of an unfamiliar Kannada name and band-limited phone audio is the worst case.

Clip 9 Majestic (rushed): only 1 hit. Models default to the English near-homophone. This is an example where an easy name fails because it collides with a common English word in the training distribution.

Conditions ranked by damage (LHR drop vs quiet baseline): traffic noise > phone call > whispered > rushed. Whispered and rushed are surprisingly survivable for short locality tokens; traffic noise causes hallucinations (clip 14 Chikkabanavara → Deepgram inserted "police" before the locality).



6. Recommendation

Match the system to the constraint:

Constraint	Pick	Why
Best accuracy on entity capture (telephony onboarding flow)	Sarvam Saarika v2.5	Highest LHR (0.55), lowest WER tier, 0.73 s latency, designed for Indian code-switch.
Lowest latency for real-time voice bots	Groq Whisper large-v3	Mean 0.43 s, p90 0.62 s, hit rate 0.50 — same accuracy tier as Sarvam at 1.7× speed. Generic model so worse on rare names.
Self-hosted, no vendor lock-in, cost control at scale	AI4Bharat IndicConformer	Open weights, India-trained, accuracy matches Sarvam, runs on a single GPU. NeMo dependencies are non-trivial — budget a day for deployment.
Default for any pipeline	Sarvam + Groq dual-write, locality post-processor	Send to both, take the higher-confidence locality after fuzzy match against the known Bangalore locality list. Catches Deepgram-style phonetic-but-wrong-spelling cases at near-zero cost.
Do not pick	Deepgram Nova-3 multi	Required as baseline by the brief, but on this dataset it is the weakest accuracy choice and not the fastest either.

7. Limitations and what I would do next

- n = 20 clips, single speaker. The numbers are directional, not statistically tight. Next step: extend to 200+ clips across 5–10 speakers, ideally with one female speaker since all five models in this batch trained predominantly on male-skewed data and the current sample doesn't probe that bias.
- Single language pair (Hindi-English Hinglish). The Vahan platform also sees Kannada, Telugu, Tamil. IndicConformer and Sarvam both claim multi-Indic support; that should be the next benchmark.
- Whisper local on Colab T4 had a 3.9 s mean latency that is dominated by Colab overhead, not the model itself. On a self-hosted L4/A10 the same model is ~0.5 s for these clip lengths; the reported latency for that system is therefore unfair and should be re-measured on dedicated hardware before any deployment decision.
- indic_whisper_fallback errored on all 20 clips on Colab. Not investigated further since IndicConformer (the primary India-tuned open-source slot) ran cleanly.
- The locality-match metric uses fuzzy + alias matching but does not currently account for cross-script phonetic equivalence (Roman ↔ Devanagari at the sound level). A more principled metric would map both reference and hypothesis to IPA before comparing — that would close the gap I noted for Deepgram in finding #3.